



EduCraft: A System for Generating Pedagogical Lecture Scripts from Long-Context Multimodal Presentations

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Abstract

Educators face substantial workload pressures, with significant time invested in preparing teaching materials. Generating high-quality lecture scripts from multimodal presentations is a particularly demanding aspect of this preparation. This paper introduces *EduCraft*, a novel system designed to automate Lecture Script Generation (LSG), addressing key difficulties such as comprehensive multimodal understanding, long-context coherence, and instructional design efficacy. *EduCraft* features a modular architecture comprising: (1) a Multimodal Input Processing pipeline for robust data extraction and association from slides; (2) a core Lecture Script Generation Engine with instruction-guided VLM and Caption+LLM workflows for pedagogical synthesis; (3) an optional Knowledge Augmentation Module using Retrieval-Augmented Generation (RAG) for enhanced factual grounding; and (4) a Model Integration and Deployment Interface supporting diverse AI models and providing a deployable API. Extensive evaluations, including human assessments and a new

automated evaluation framework, demonstrate that *EduCraft* significantly outperforms strong baselines and teacher-refined scripts in producing coherent, readable, and pedagogically sound lecture scripts. By effectively tackling core LSG challenges, *EduCraft* offers a practical, configurable solution to reduce educator workload and enhance educational content creation. We open-source *EduCraft* at <https://github.com/wyuc/EduCraft>.

CCS Concepts

• **Applied computing** → **E-learning; Interactive learning environments**; • **Computing methodologies** → *Natural language generation; Reasoning about belief and knowledge*.

Keywords

Lecture Script Generation, Long-Context LLMs, Vision-Language Models, Retrieval-Augmented Generation

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1 Introduction

Selecting, applying, and even creating technological tools and instructional resources to support students' active learning is one of the key standards for accomplished teaching¹. Before the National Board for Professional Teaching Standards (NBPTS) formally announced this requirement in 1989 [14], generations of educators had already devoted immeasurable labor to curriculum preparation [9, 19], as a substantial course preparatory period often involves the meticulous and time-consuming crafting of teaching materials from diverse sources. Among these activities, the development of comprehensive lecture scripts from visual aids like slides emerges as a particularly demanding endeavor, requiring educators to painstakingly synthesize complex information, structure coherent narratives, and ensure pedagogical effectiveness under these significant time constraints. The technical pursuit of automating this labor-intensive process is known as *Lecture Script Generation* (LSG). This task involves the automated transformation of slides into a structured, pedagogically sound narrative script suitable for oral delivery. Such scripts are vital not only for direct classroom instruction but also as foundational assets for broader educational resources, including Massive Open Online Courses (MOOCs) [15] and content for Intelligent Tutoring Systems (ITS) [23, 25], thereby playing a crucial role in disseminating high-quality education.

The emergence of powerful Large Language Models (LLMs) presents a significant opportunity to automate key aspects of LSG, potentially easing this preparatory burden. Indeed, state-of-the-art models such as GPT-4V [11] and Gemini [17] showcase remarkable generative capabilities in multimodal understanding and handling long contexts. However, bridging the gap between these general capabilities and effective solutions for the specific technical challenges inherent in LSG reveals significant hurdles from both algorithmic performance and engineering deployment perspectives:

- **Formidable Obstacles for Algorithmic Performance.** The LSG task naturally lies at the intersection of multiple weak capabilities of existing large models: (1) *Comprehensive Multimodal Understanding*. Despite strong unimodal processing, accurately synthesizing information from visual elements (e.g., images, diagrams, charts) and slide text, especially their intricate interplay, remains a primary obstacle. (2) *Maintaining Long-Context Coherence*. Though adept with longer sequences, ensuring narrative consistency, logical flow, and thematic connections across slides remains a persistent issue, as models struggle with long-context dependencies (Figure 1) [18, 21]. (3) *Instructional Design for Educational Efficacy*. Beyond understanding and coherence, even with sophisticated generation, crafting pedagogically effective scripts—translating content into clear, engaging, and instructionally sound sequences—is a nuanced task challenging for current models [12].

- **Pressing Needs for Engineering Deployment.** To be effectively used in complex real-world educational scenarios, this system must also possess the following characteristics: (1) *Flexible and Robust System Design*. Despite increasing model versatility, engineering systems to reliably process diverse slide formats, integrate various foundation models, and adapt to user/institutional needs remains practically difficult. (2) *Ensuring Factual Accuracy and Knowledge Integration*. Despite impressive recall, integrating

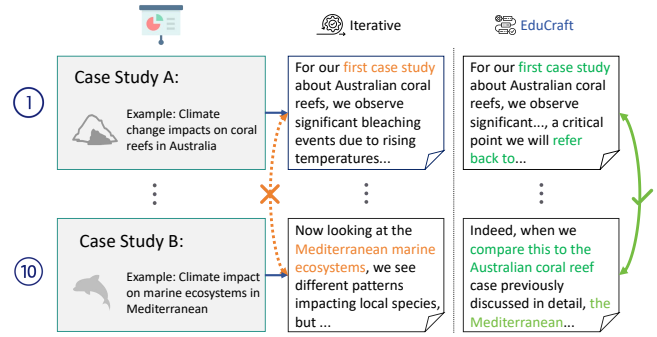


Figure 1: Challenge of long-range coherence in LSG: Traditional generation methods, exemplified by iterative approaches (middle), struggle to link distant concepts (e.g., Slide 1 to Slide 10), a gap EduCraft (right) aims to bridge.

mechanisms like RAG [3, 6] to ground content factually, provide information, and enable domain customization remains a critical hurdle for deployment. (3) *Practical Deployment and Usability*. Despite powerful models, engineering effective, efficient, scalable systems that seamlessly merge with existing educational platforms and workflows remains a substantial practical challenge.

To address these gaps, we develop **EduCraft** (Section 3), a modular system for high-quality lecture script generation. Its *Multimodal Input Processing* pipeline tackles Comprehensive Multimodal Understanding by extracting and associating diverse slide elements. The core *Lecture Script Generation Engine*, with VLM and Caption+LLM workflows, addresses Long-Context Coherence and Instructional Design for Educational Efficacy through holistic interpretation and pedagogical synthesis. An optional *Knowledge Augmentation Module* (RAG) targets Ensuring Factual Accuracy. Finally, the *Model Integration and Deployment Interface* promotes Flexible and Robust System Design and Practical Deployment and Usability via broad model compatibility and a deployment API.

Through extensive experiments detailed in Section 4, EduCraft's effectiveness was validated using both automated metrics and human evaluations. Human judges found EduCraft's scripts to be significantly superior in consistency, readability, and coherence when compared to iterative baselines and even teacher-refined versions. Automated evaluations further confirmed that EduCraft's methodologies consistently surpass baseline approaches in generating high-quality, pedagogically sound lecture scripts. These results highlight EduCraft's capability to effectively tackle the core LSG challenges and produce outputs that approach human standards.

Our main contributions are: (1) A novel methodology embodied in the EduCraft system, designed to generate high-quality, pedagogically sound lecture scripts by addressing core LSG challenges (Section 3). (2) The EduCraft system itself, a practical and configurable platform supporting diverse AI models and generation strategies (including VLM and Caption+LLM workflows with optional RAG) to reduce educator workload and facilitate content repurposing (Section 3). (3) A comprehensive evaluation framework, including human assessments and a novel set of automated metrics, demonstrating EduCraft's superior performance and the efficacy of its design in producing high-quality lecture scripts (Section 4).

¹<https://www.nbpts.org/>

2 Preliminaries

2.1 Related Work

Long Context Vision Language Models (LC-VLMs). LC-VLMs like GPT-4V [11] and Gemini [17] integrate vision encoders with extended-context LLMs to process multimodal content across numerous slides. While relevant to LSG, ensuring narrative coherence and factual grounding over long sequences remains a significant and ongoing challenge for current models [18, 21].

Automated Lecture Script Generation. Early efforts in LSG focused on slide summarization [13] and note generation [22], but lacked deep multimodal understanding. While newer models enable transcript creation [10], challenges persist in maintaining long-range coherence and pedagogical depth [12, 18, 21]. Systems like Slide2Lecture [26] and GenMentor [20] share some LSG goals but do not specifically address comprehensive script creation with the required pedagogical depth and instructional quality.

2.2 Problem Formulation

Given a presentation $P = \{s_1, \dots, s_N\}$, where each slide s_i contains visual (V_i) and textual (T_i) elements, the goal is to generate a corresponding lecture script $S = \{L_1, \dots, L_N\}$. Each segment L_i is the script generated for slide s_i , narrating the multimodal information from it as a coherent and pedagogically sound narrative for delivery. This script should effectively bridge the gap between slide content and an engaging, instructive oral presentation.

3 EduCraft

EduCraft tackles challenges from Section 1 to convert slide presentations into pedagogically sound lecture scripts. Its modular architecture (Figure 2) offers flexible, targeted solutions through core components and optional augmentations.

3.1 Core Workflow

This subsection details the algorithmic steps EduCraft employs to transform multimodal presentations into coherent lecture scripts.

3.1.1 Multimodal Input Processing. To enable a deep understanding of the inherently multimodal slide content, this initial stage implements a robust pipeline to extract and structure data from source presentations. **Image Sequence Acquisition** involves converting each slide into a high-fidelity raster image, capturing all visual information such as diagrams, charts, and spatial layouts. **Textual Content Extraction** utilizes a hierarchical parsing mechanism to accurately extract textual content (T_i) from various file formats, preserving its structure. This extracted text is considered the ground truth for on-slide textual information. **Text-Visual Association** is a critical step that links the ground-truth extracted text (T_i) to its corresponding slide image (V_i). This association ensures that the generation model has access to accurate textual information, which is prioritized over potentially error-prone VLM-based image-to-text recognition, and aids in grounding visual elements by linking them to their intended textual descriptors. Optionally, for workflows requiring explicit textual descriptions of visual content, an additional **Visual Caption Generation** step can be employed. Here, a vision-language model (VLM) or a dedicated image-to-text model

processes the slide image (V_i) to produce a concise descriptive caption (C_i). This stage outputs a comprehensive multimodal representation for each slide, comprising the slide image (V_i), ground-truth extracted textual content (T_i), and visual caption (C_i).

3.1.2 Lecture Script Generation Core. This core engine transforms the processed multimodal data into a coherent, engaging, and pedagogically effective lecture script. It supports two primary operational workflows: (1) a **direct Vision Language Model (VLM) workflow**, where a VLM holistically processes the slide image (V_i) and its associated ground-truth textual content (T_i) to generate the script segment for that slide; and (2) a **two-stage Caption+LLM workflow**, which utilizes the generated visual caption (C_i) and the extracted textual content (T_i). These two textual streams are then fed into a Large Language Model (LLM) for script generation. This decoupled approach allows for flexibility, such as using a specialized vision model for captioning and a powerful LLM for narrative synthesis, potentially optimizing for efficiency, cost, or model capabilities. Both workflows operate through the following sequential process, designed to emulate expert human lecturer scriptwriting:

1. Holistic Slide Interpretation and Prioritization. The system first analyzes all slide images (V_i) (or visual caption (C_i)) and extracts text (T_i) to understand course structure and establish a global narrative context, which is foundational for maintaining *long-range coherence*. For each slide, V_i informs the understanding of design and layout, while the ground-truth T_i ensures textual accuracy, correcting potential VLM errors in reading on-slide text.

2. Pedagogical Content Transformation. Slide information is transformed into pedagogically effective content. Visuals are converted to core concepts and teaching points, avoiding direct description using conceptual language, metaphors, or scenarization (e.g., a process diagram becomes an explanation of a cyclical process). Key informational content (e.g., academic works, models, data) is integrated, while decorative elements are ignored. Mathematical content is explained conceptually, focusing on meaning and approach over verbatim recital or step-by-step calculation.

3. Fluent and Engaging Script Synthesis. Finally, this transformed content is synthesized into a fluent, coherent, and engaging lecture script segment (L_i) for each slide. Scripts are logically structured, providing value-added transformation of slide content by reorganizing language rather than mere repetition. Crucially, this stage focuses on achieving *long-range coherence* by weaving individual script segments into a unified narrative, referencing concepts from earlier slides where appropriate, maintaining thematic consistency, and ensuring smooth transitions that bridge topics across the entire lecture. The system adheres to the teaching rhythm, ensures natural slide transitions, and optimizes language for clear, colloquial, and professional oral delivery (avoiding mixed-language or parenthetical interruptions). It incorporates techniques to enhance educational value (e.g., emphasis, guiding questions) and liveliness, adhering to established principles for overall script construction.

This multi-stage synthesis process, driven by detailed pedagogical instructions, is the primary pathway for script generation.

3.2 System Architecture and Implementation

This subsection describes the supporting modules and system-level design choices that ensure EduCraft is robust and flexible.

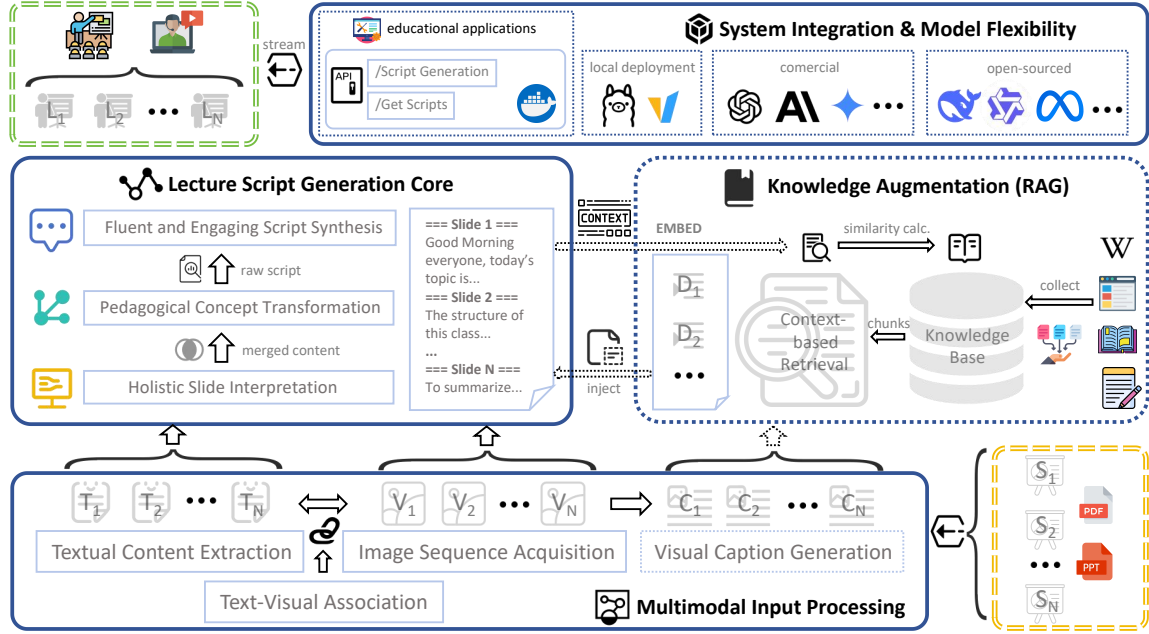


Figure 2: The modular architecture of EduCraft. Information flows from the input presentation (PPT or PDF) through the Multimodal Presentation Ingestion Framework (extracting image V_i , text T_i , and optional caption C_i), into the Pedagogical Narrative Synthesis Engine (including 3 steps) for script generation (L_i). The system is enhanced by optional Knowledge Augmentation (RAG), flexible Model Integration capabilities, and an API Container for deployment and application integration.

3.2.1 Knowledge Augmentation Module (RAG). To address scenarios requiring enhanced factual accuracy or incorporating external information (e.g., for specialized topics or when using models with knowledge cutoffs), EduCraft includes a Knowledge Augmentation module based on Retrieval-Augmented Generation (RAG). This module can operate with the Lecture Script Generation Core. When activated, it queries an external knowledge base (e.g., domain-specific texts, curated web sources) using context derived from the current slide. The retrieved relevant information, D_i , is then integrated into the Instructional Concept Mapping or Coherent Script Synthesis stages. This allows EduCraft to enhance factual accuracy, provide deeper context, and tailor scripts to informational needs beyond the base model’s knowledge.

3.2.2 Model Integration and Deployment Interface. To ensure EduCraft is practical for real-world use, this component addresses system-level aspects of model choice and application integration. **Flexible Model Support.** The system is designed to interface with a wide array of foundation models. It supports leading proprietary APIs (e.g., from Anthropic, Google, OpenAI) as well as open-source models deployable locally (e.g., via Ollama, vLLM). This enables users to select models based on specific requirements, such as performance, cost, or data privacy. The optional RAG module (Section 3.2.1) further enhances this flexibility by providing a means to augment the capabilities of any chosen model.

Deployment API. For seamless integration into various educational platforms and applications (e.g., Learning Management Systems), EduCraft is packaged as a configurable and deployable container. This container exposes a streaming API for lecture script

generation, designed to support dynamic content creation and delivery. The API allows configuration of model selection, generation style, and activation of the RAG module. The structured script output simplifies downstream use and integration.

This layered architecture ensures EduCraft’s generation capabilities are accessible and deployable in diverse educational contexts.

4 Experiment

To evaluate EduCraft, we conducted comprehensive experiments comparing it against alternative workflows and assessing its utility via automated metrics and human judgment.

4.1 Human Evaluation

Human judgment is crucial for assessing nuances like pedagogical effectiveness, which automated metrics may not fully capture. We employed a comparative ranking methodology, known to be more reliable than absolute scoring for discerning relative quality in human evaluations [4]. To capture key aspects of a high-quality lecture script suitable for delivery, we focused on three established dimensions for text evaluation:

- **Consistency:** Ensuring factual alignment within the script, logical flow of arguments without contradictions, and adherence to the content presented on the source slides [5, 16].
- **Readability:** Assessing the clarity, conciseness, and linguistic quality of the script, focusing on its suitability for oral delivery and ease of audience comprehension [24].

• **Coherence:** Evaluating how well individual utterances and ideas are connected, forming a well-structured and unified narrative that flows logically from one point to the next [1].

These metrics were chosen as they directly impact the pedagogical effectiveness and overall comprehensibility of a lecture script.

Evaluation Procedure. For human evaluation, we selected 20 presentations from 4 distinct university-level courses on the MAIC platform [23], averaging 16 slides each (320 slides total) and covering diverse academic disciplines. Utilizing these presentations, which represent actual teaching scenarios from MAIC, was crucial for a realistic evaluation. This setup allowed for direct comparison against two authentic baseline conditions derived from MAIC’s functionalities: scripts generated by its standard iterative, slide-by-slide method [26], and scripts that had been subsequently refined by educators using the platform’s tools. These MAIC-based scenarios served as our primary baselines against EduCraft:

(1) *Iterative Baseline:* Scripts generated using MAIC’s standard iterative, slide-by-slide method.

(2) *Teacher Refined:* Scripts from the Iterative Baseline, manually edited and improved by teachers using MAIC’s refinement features.

(3) *EduCraft:* Scripts generated by our proposed EduCraft system using the Claude 3.7 Sonnet VLM workflow.

For each of the 20 presentations, the three script versions were presented in a randomized order to annotators for comparative ranking on the three aforementioned quality dimensions.

Annotation Process and Quality Control. The evaluation was conducted by a group of 10 annotators, comprising graduate students who were familiar with the subject matter of the presentations. Annotators were provided with a detailed guideline document that clearly defined Consistency, Readability, and Coherence, including positive and negative examples for each metric. The guidelines also specified the ranking task: for each criterion and each presentation, annotators were asked to rank the three script versions from best (3) to worst (1), and ties were not permitted. To ensure understanding and calibration, a pilot annotation round was conducted by a separate group of 3 graduate students with similar backgrounds, where they ranked the full set of 20 presentations; this was followed by a group discussion and refinement of the guidelines based on their feedback. For the main study, the 10 annotators (distinct from the pilot group) then evaluated all 20 presentations using the finalized guidelines. The final rank for each script version on each criterion was determined by averaging the scores from these 10 annotators.

Results. The human evaluation results, summarized in Figure 3, demonstrate EduCraft’s strong performance. EduCraft (average rank of 2.41) significantly outperformed the Iterative Baseline (1.47) and also surpassed the Teacher Refined version (2.12) across all three criteria: Consistency (EduCraft: 2.44 vs. Iterative: 1.51, Refined: 2.04), Readability (EduCraft: 2.37 vs. Iterative: 1.48, Refined: 2.16), and Coherence (EduCraft: 2.44 vs. Iterative: 1.31, Refined: 2.25). This suggests EduCraft’s holistic, instruction-guided synthesis produces higher quality scripts. The advantage over the Teacher Refined versions might be attributed to EduCraft’s fresh, holistic generation process, whereas the teacher refinements were based on improving the Iterative Baseline scripts, potentially constraining the scope of revisions and inheriting limitations from that initial, less optimal output. The strong performance in human evaluations, indicating

that EduCraft can produce scripts preferred over even teacher-refined ones, underscores its capability to generate content of high pedagogical quality.



Figure 3: Human Evaluation Ranking Results (Average Scores). Higher is better (Max 3.0). The results show that EduCraft outperforms both the Iterative Baseline and Teacher Refined versions across all evaluation metrics.

4.2 Automated Evaluation

Automated metrics serve as a crucial complement to human evaluation. While human judgment is vital for assessing perceived quality and pedagogical effectiveness, automated metrics provide scalable, cost-effective, and reproducible assessments of specific linguistic and structural characteristics.

Experimental Setup. Our automated evaluation utilized a diverse dataset of 20 university presentations (10 English, 10 Chinese), with an average length of 13.6 slides per presentation (272 slides total). The selection aimed to ensure broad coverage of academic topics, encompassing disciplines across the Humanities, Social Sciences, STEM (including computer science, engineering, and physical sciences), and applied fields to reflect a realistic range of university-level content. We compared three VLM-based workflows: an *Iterative Baseline* (slide-by-slide, limited context, inspired by [26]), a *Direct Prompt Baseline* (holistic VLM, minimal prompt), and *EduCraft’s VLM workflow* (holistic VLM, detailed guidance). Models tested included Qwen-2.5-VL-72B and GPT-4o (2024-11-20 version).

Evaluation Metrics. To comprehensively evaluate the generated scripts, we propose a new evaluation system based on Lim et al. [7]. Slide-level quality was assessed using metrics adapted from this system, grounded in Mayer’s Cognitive Theory of Multimedia Learning [8]: Content Relevance (CR), Expressive Clarity (EC), Logical Structure (LS), and Audience Engagement (AE). Global coherence was assessed by Claude 3.7 Sonnet on five dimensions: Narrative Flow (NF), Information Hierarchy (IH), Conceptual Integration (CI), Thematic Consistency (TC), and Cross-referencing (CRef), drawing from discourse analysis and instructional design principles [2]. Table 1 reports these slide and global scores, along with aggregates: Avg. Slide Score (Avg-S), Global Score (Glo), and their average, the Combined Score (Comb).

Results and Analysis. Table 1 reveals key findings regarding the EduCraft VLM workflow. EduCraft’s VLM workflow consistently

Table 1: Automated Evaluation Results: Performance across Slide Dimensions (Content Relevance (CR), Expressive Clarity (EC), Logical Structure (LS), Audience Engagement (AE)), Global Dimensions (Narrative Flow (NF), Information Hierarchy (IH), Conceptual Integration (CI), Thematic Consistency (TC), Cross-referencing (CRef)), and Overall Scores (Avg. Slide Score (Avg-S), Global Score (Glo), Combined Score (Comb)). Best scores are in bold. GPT-4o refers to gpt-4o-2024-11-20. EduCraft’s VLM workflow generally demonstrates superior performance over iterative and direct prompt baselines across these metrics.

Model	Slide Dimensions				Global Dimensions					Overall Scores		
	CR	EC	LS	AE	NF	IH	CI	TC	CRef	Avg-S	Glo	Comb
<i>Iterative Baseline</i>												
Qwen2.5-VL-72B	4.19	4.14	4.11	2.46	3.80	3.85	3.65	4.65	2.90	3.73	3.77	3.75
GPT-4o	4.13	4.16	4.06	2.36	3.80	3.80	3.60	4.55	2.85	3.68	3.72	3.70
<i>Direct Prompt Baseline</i>												
Qwen2.5-VL-72B	4.12	4.20	4.09	2.48	3.85	3.90	3.40	4.80	2.75	3.72	3.74	3.73
GPT-4o	4.11	4.20	4.05	2.39	3.85	4.05	3.85	4.80	2.85	3.69	3.88	3.78
<i>EduCraft: VLM</i>												
Qwen2.5-VL-72B	4.20	4.23	4.14	2.49	3.85	4.00	3.85	4.85	2.80	3.77	3.87	3.82
GPT-4o	4.16	4.22	4.11	2.48	4.00	3.95	4.00	4.80	3.15	3.74	3.98	3.86

Table 2: Automated Evaluation Overall Scores for EduCraft Caption+LLM Workflow. Scores are on a 1-5 scale.

Model (LLM)	Avg-S	Glo	Comb
DeepSeek-V3	3.65	4.14	3.89
Qwen2.5-72B	3.70	3.79	3.74

outperformed both baselines with the same models. For instance, using GPT-4o (2024-11-20 version), EduCraft VLM achieved a Combined Score of 3.86, compared to 3.78 for the Direct Prompt baseline and 3.70 for the Iterative baseline. Similarly, with Qwen2.5-VL-72B, EduCraft VLM scored 3.82, surpassing the Direct Prompt (3.73) and Iterative (3.75) baselines. These results highlight the importance of EduCraft’s comprehensive instructional schema for enhancing script quality and coherence. Overall, automated metrics demonstrate that EduCraft’s VLM workflow generally surpasses baselines in both slide-level dimensions and global coherence measures.

4.3 EduCraft Caption+LLM Workflow Evaluation

As an alternative approach, EduCraft also supports a two-stage Caption+LLM workflow. This design offers flexibility, allowing for the use of specialized vision models for captioning and powerful Large Language Models for narrative synthesis, potentially optimizing for cost, efficiency, or specific language capabilities. The performance of this workflow was evaluated using GPT-4o-mini for captioning, combined with DeepSeek-V3 and Qwen2.5-72B for script generation. The key overall scores are summarized in Table 2.

The results presented in Table 2 indicate that the Caption+LLM workflow can achieve strong performance. For instance, DeepSeek-V3 achieved a notable Global Score (Glo) of 4.14 and a Combined Score (Comb) of 3.89. Similarly, Qwen2.5-72B obtained a Combined

Table 3: Impact of RAG Integration on Overall Combined Score (S_{Comb}). Baseline scores for EduCraft VLM (No RAG) are from Table 1. GPT-4o refers to gpt-4o-2024-11-20.

Model	Base VLM	VLM+RAG	Δ
Qwen2.5-VL-72B	3.82	3.75	−0.07
GPT-4o	3.86	3.83	−0.03

Score of 3.74. These findings validate the Caption+LLM approach as a viable and effective option within the EduCraft framework.

4.4 Impact of RAG Integration

To further investigate RAG’s potential for enhancing factual grounding and contextual relevance, we explored its integration as an optional component within EduCraft’s core VLM workflow. This experiment aimed to assess the impact of incorporating external knowledge from a Wikipedia-based knowledge base on the quality of the generated lecture scripts.

Experimental Setup for RAG. We constructed a knowledge base by automatically extracting relevant topics from presentation content, fetching corresponding articles from Wikipedia, and then cleaning and segmenting these articles into manageable chunks suitable for retrieval. RAG was implemented within the VLM workflow by retrieving relevant chunks based on slide content and incorporating this information into the generation workflow. We evaluated this RAG-enabled VLM configuration (denoted EduCraft VLM+RAG) using the same 20 presentations and foundation models (Qwen-2.5-VL-72B, GPT-4o (2024-11-20 version)) employed in the primary automated evaluation.

Results and Analysis. As Table 3 indicates, RAG integration led to a slight decrease in overall Combined Scores. However, the utility of RAG is better understood through qualitative analysis. Figure 4 illustrates this trade-off: the "Good Case" example shows CR enhancement via factual enrichment (e.g., Atacama Giant details),

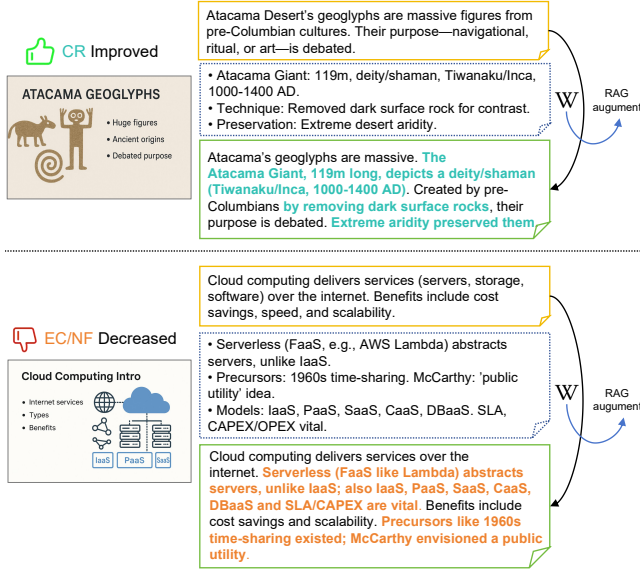


Figure 4: Qualitative examples illustrating the context-dependent impact of RAG integration. (Up) RAG enhances Content Relevance (CR) by adding factual details (e.g., Atacama Giant specifics). (Down) RAG can decrease Expressive Clarity (EC) or Narrative Flow (NF) by introducing overly technical or tangential information (e.g., Cloud Computing acronyms). Bold text highlights RAG-influenced additions.

while the "Bad Case" example demonstrates potential EC/NF degradation from overly technical or tangential information.

These examples highlight that RAG's impact is context-dependent. It can enrich content but may also harm clarity if not carefully applied. Thus, EduCraft offers RAG as an optional feature, ideal for specific content augmentation needs rather than default use.

4.5 Ablation Study

To better understand the individual contributions of its core architectural elements, we conducted a focused ablation study on EduCraft's VLM workflow using GPT-4o (2024-11-20 version). This study systematically ablated three key modules of our Pedagogical Narrative Synthesis Engine (see Section 3): the *Coherent Narrative Generation* module (responsible for aspects like tone and engagement), the *Pedagogical Concept Mapping* module (which links content to learning objectives and structures explanations), and the *Multimodal Content Grounding* module (which explicitly references and explains visual slide elements). The impact on the overall Combined Score is detailed in Table 4).

The results affirm the value of each ablated module. Removing any of these components led to a decrease in performance compared to the full EduCraft VLM workflow's baseline score of 3.86. Notably, ablating the Multimodal Content Grounding module resulted in the most significant drop ($\Delta = -0.11$). This highlights the critical importance of this module in explicitly guiding the VLM to connect its narrative to the visual information presented on the slides, ensuring that the script and visuals are synergistically aligned. While

Table 4: Ablation Study: Impact of Removing Guidance Components (EduCraft VLM with GPT-4o (2024-11-20 version))

Configuration	S_{Comb}	Δ
Full EduCraft VLM Workflow	3.86	–
w/o Narrative Style	3.80	–0.06
w/o Pedagogical Mapping	3.81	–0.05
w/o Multimodal Grounding	3.75	–0.11

removing the Coherent Narrative Generation module also impacted scores ($\Delta = -0.06$), and ablation of the Pedagogical Concept Mapping module had a similar effect ($\Delta = -0.05$), the stronger impact of removing Multimodal Content Grounding underscores its central role in generating effective lecture scripts from visually rich presentations. These findings collectively validate the integrated design of EduCraft's Pedagogical Narrative Synthesis Engine.

5 Conclusion

This paper introduces EduCraft, a system designed to automate the generation of high-quality lecture scripts from multimodal slides, addressing a key challenge in educator workload. EduCraft employs a modular architecture featuring multimodal input processing, an instruction-guided lecture script generation engine, and optional knowledge augmentation via RAG. Our comprehensive evaluations, incorporating both automated metrics and human assessments, demonstrated that EduCraft significantly outperforms baseline methods and produces scripts comparable in quality to those refined by educators. The system's core contributions lie in its novel methodology for creating pedagogically sound scripts, its practical and configurable design supporting diverse AI models, and strong empirical validation of its effectiveness. Its design not only addresses the technical complexities of LSG but also offers a practical solution for educators. This work therefore, contributes a valuable tool for enhancing educational material development and delivery. By streamlining the script generation process, EduCraft empowers educators to focus more on teaching and less on material preparation. The system's adaptability to various AI models also ensures its long-term relevance and utility in the evolving landscape of educational technology and new AI-assisted learning tools.

Future work will focus on:

- **Enhanced RAG Integration:** Investigating sophisticated retrieval and integration strategies to improve factual accuracy and contextual relevance, preserving narrative flow.
- **Interactive Refinement Features:** Developing tools that allow educators to collaboratively refine and personalize generated scripts, providing more granular control over the final output.
- **Personalized Script Generation:** Exploring advanced techniques to meticulously tailor lecture scripts to diverse students' profiles and specific learning objectives.

EduCraft represents a significant step towards bridging advanced AI capabilities and the practical needs of modern education, aiming to enhance content creation and student learning outcomes.

6 GenAI Usage Disclosure

Generative AI (GenAI) tools, particularly Vision Language Models (VLMs) and Large Language Models (LLMs), were utilized at multiple stages of this research. For manuscript preparation, an LLM-based writing assistant provided support for refining grammar, enhancing clarity, and improving conciseness. The conceptualization of the research, the design of the EduCraft system, the experimental methodology, and the interpretation of all findings represent the original intellectual contributions of the human authors exclusively; no section of this paper was directly composed by an LLM.

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